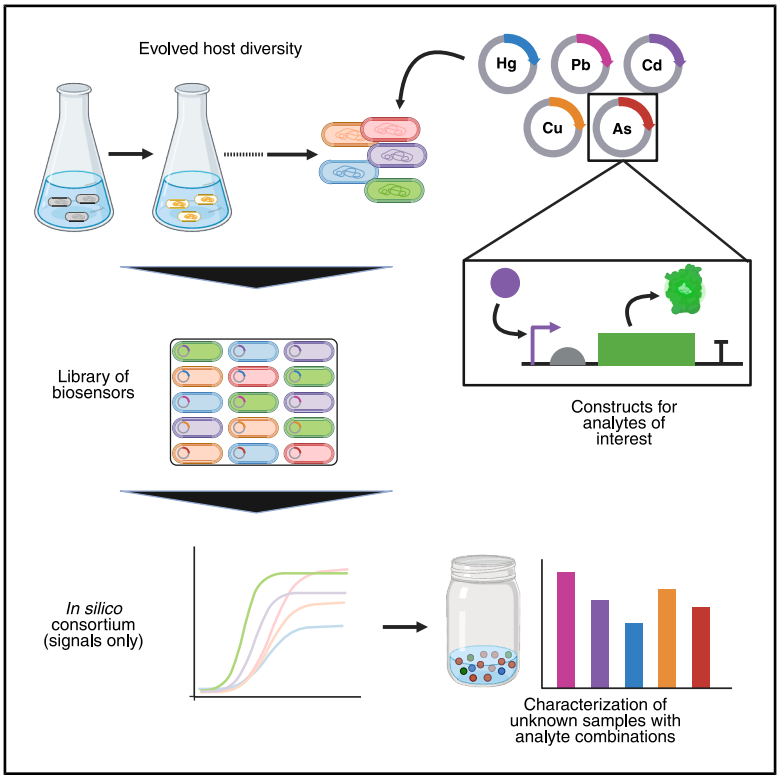
Article

Cell Systems

# A diversified consortium of engineered biosensors enables multi-target quantification

## Graphical abstract




## Highlights

- Adaptive laboratory evolution generates biosensor chassis robust to high salinity
- Microfluidic screening reveals diversity across engineered synthetic biosensor variants
- Biosensor signals are combined in an *in silico* consortium to train machine learning model
- Model outperforms single-target "lock-and-key" biosensors in combinatorial quantification

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## In brief

Whole-cell biosensors are typically engineered for single-target detection under controlled laboratory conditions, and this limits their deployability. Chiang et al. introduce an *in silico* consortium sensing paradigm that leverages response diversity rather than relying on lock-and-key specificity. Combining adaptive laboratory evolution, microfluidic screening, and supervised machine learning, the authors integrate cross-reactive biosensor responses to quantify multiple heavy metals in seawater. This consortium model paradigm establishes a generalizable path for deployable microbial sensing.

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# A diversified consortium of engineered biosensors enables multi-target quantification

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## SUMMARY

Whole-cell biosensors (WCBs) offer rapid, cost-effective monitoring of environmental contamination. Efforts to optimize detection of isolated target analytes under laboratory conditions have achieved vastly improved performance and set the stage for WCB deployment in complex environments. We propose a framework that leverages cross-reactivity of single-target WCBs to quantify multiple targets using supervised machine learning. Specifically, we engineer six sensors for heavy metal contaminants in laboratory *E. coli*. We then evolve the strain to generate five chassis with improved growth in seawater conditions and transform them with the sensors to create a set of 30 variants. The variant responses are characterized with microfluidics, revealing significant diversity. Leveraging this diversity, we combinatorially quantify multiple analytes with a machine learning model that takes an *in silico* consortium of response inputs and outperforms single-target WCBs in over 90% of test samples. These results form a generalizable framework that facilitates WCB translation toward settings beyond the laboratory. A record of this paper's transparent peer review process is included in the supplemental information.

## INTRODUCTION

Human industrial and agricultural activities have resulted in considerable waste and toxic byproducts that have left marine environments with increasing levels of pollutants such as heavy metals.[1–3] As they accumulate, heavy metals become toxic, harming biodiversity[3,4] and presenting devastating neurological and physiological risks to human health.[4–6] As a result, cost-effective biomonitoring solutions that are robust to high-salinity conditions while capable of frequent and rapid monitoring are essential to facilitating pollutant mitigation and reduction efforts.

Whole-cell biosensors (WCBs) have gained significant traction in the detection of target analytes, informing biological solutions in human or environmental health,[7–10] agriculture,[11] or food safety.[12] Though chemical or physical detection methods exist for many analytes—such as mass spectrometry or high-performance liquid chromatography[13,14]—these assays can be expensive, time-consuming, and salinity-sensitive, while also requiring experienced personnel or complex equipment.[15] WCBs, on the other hand, offer affordable, versatile, rapid, and simple readouts to address these needs.[16,17] Furthermore, they can be engineered to be linked to remedial actions such as therapeutic payload release[18,19] or toxin neutralization.[20]

WCBs have employed a variety of sensing approaches, including DNA-based,[21,22] RNA-based,[22,23] transcription factor-based,[7,9,24–26] and optical approaches.[27,28] With the rise of synthetic biology and genetic engineering precision,[29–31] transcription factor-based sensors have dominated the WCB space.[32,33] Engineering efforts have resulted in precise biosensors for the detection of compounds including heavy metals,[7,9,34–36] caffeine,[24] aromatics,[37,38] and waterborne pathogens.[39] More recently, evolutionary and computational approaches have expanded the repertoire of possible targets and improved the precision of existing biosensors.[40–45]

However, transcription factor-based WCB development has focused largely on engineering and tuning strains in a lock-and-key approach to detect single targets within a controlled laboratory environment.[46,47] While these advances have vastly


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improved WCB capabilities, their focus on single-target sensing in well-characterized and controlled environments limits their deployability in complex environments, leaving a gap in realizing WCBs with field potential.[48,49] In fact, single-target biosensors often suffer from cross-reactivity, preventing the simultaneous detection of multiple biomarkers in a sample.[49–52] Furthermore, WCBs characterized under laboratory settings face difficulties in survival, long-term stability, and signal interference[53] under environmental stressors, including temperature extremes, pH, and osmolarity, as well as exposure to toxins and oxidative compounds.[54]

Here, instead of optimizing single-target WCBs, we focus on sensing multiple target analytes in mixed samples under high-salinity seawater conditions. Natural olfactory systems, capable of interpreting signal patterns collectively transmitted through hundreds of receptors, have inspired analogous approaches in the biosensor space.[47,55–57] The development of electronic noses[58] and tongues,[59] paired with the rise of machine learning techniques, has motivated efforts to integrate cross-reactive biosensors into "fingerprint" readouts to distinguish complex mixtures. A consortium-based readout circumvents the need for engineering microbial precision while expanding the potential of current sensors to enable combinatorial detection of multiple targets. To this end, we harness adaptive laboratory evolution (ALE) with the well-characterized *E. coli* K-12 MG1655 laboratory strain to efficiently generate a library of diverse biosensor signals to five target heavy metals of concern in polluted seawater samples, circumventing the problem of environmental fitness defects and the need to characterize a new chassis altogether, without requiring *de novo* information about microbial stress tolerance. To more fully capture the phenotypic diversity of the evolved library, we develop a microfluidic screening tool, an affordable and versatile microenvironment that can elucidate WCB dynamic behavior in continuous culture with high temporal resolution.[7,60,61] Our modular microfluidic design allows parallel screening of multiple strains or conditions in a single experiment. Exploiting our WCB diversity, we expose a set of strains to approximately 500 combinations of the five heavy metals in seawater to extract their signals into an *in silico* consortium to train a simple supervised learning regression model to quantify the metals, using the consortium response as a fingerprint. Instead of relying on single-target WCB responses, using the collective response yields estimations of heavy metal concentrations that outperform those of single-target biosensors, which are confounded by signal cross-reactivity.

An *in silico* consortium paradigm of WCBs opens a path toward field deployability by shifting the engineering approach from optimization of single-target WCBs toward maximization of data collected from WCBs. The fingerprint biosensor model can be transposed to a myriad of diagnostic applications that require quantification of multiple biomarkers under stressful environmental conditions.

## RESULTS

### Engineered transcription factor-based constructs in *E. coli* detect heavy metals of interest

We sought to engineer and characterize conventional transcription factor-based constructs to monitor five heavy metals relevant to marine environments[2,3]: arsenic, mercury, copper, lead, and cadmium. To this end, six sensor constructs were engineered, each consisting of a metal-inducible promoter that drives green fluorescent protein (GFP) expression and, in some cases, its own regulating transcription factor (Figure 1A; Table S1). Each construct was designed to be specific to a certain metal and to produce a fluorescent output corresponding to the metal concentration present.

All sensor constructs were placed on a medium-copy p15A origin plasmid with a kanamycin resistance gene. The native *E. coli* $P_{arsR}$ promoter was selected for arsenic detection. This promoter has previously been characterized and shown to respond to arsenic through the action of the transcriptional repressor protein ArsR, which binds to the promoter region and inhibits transcription in the absence of arsenic.[62] Similarly, the native *E. coli* $P_{cadC}$ promoter was selected for sensing cadmium, as it has previously been characterized to behave similarly to the arsR system, via the transcriptional repressor protein CadC[63]. For detection of mercury, we selected the well-characterized MerR transcriptional activator with its corresponding bidirectional promoter. While merR is naturally found in a variety of Gram-negative bacteria and plasmids on the transposon Tn21,[64] it is not naturally found in the MG1655 strain. In addition, the promoters $P_{cusC}$, which is sensitive to copper,[65] and $P_{zraP}$, which is sensitive to lead,[66] are natively present on the MG1655 genome and were previously found responsive without overexpression of additional transcriptional factors. Promoter $P_{zntA}$, responsive to cadmium and lead,[67] was also included, given its well-documented sensor performance.

Sensor construct performance was initially verified in batch in wild-type *E. coli* K-12 MG1655 in minimal media, with dose-response curves measured for each strain using the Hill equation (Figure 1B). A detectable level of fluorescence was measured as a function of concentration for all sensors besides cadC, which displayed little to no correspondence between fluorescence and concentration. Moreover, several sensors arrest growth before the GFP response is saturated, indicating a limited operational range for each of the sensors beyond which sensor output may become unreliable (Figure S1). The zntA sensor responded to both cadmium and lead as intended, but its cross-reactivity for the two metals revealed a major weakness of WCBs manufactured to detect isolated target analytes: cross-reactivity across multiple analytes can lead to confounding signals in mixed samples.

### Multiplexed microfluidics enables efficient dynamic screening and characterization

For further characterization, we developed a versatile, cost-effective microfluidic screening system with modular design elements to enable simultaneous testing of multiple microbial strains as well as varied conditions in parallel. WCBs are typically characterized using single time-point approaches, often masking dynamic behaviors, which can carry more information on WCB response over time and measure responses to media changes in real time.[68–70]

Central to our platform are the hydrodynamic (HD) cell trapping regions, which are easily interchangeable with different architectures, depending on end user needs using our microfabrication process.[71,72] Hydrodynamically trapping cells in a

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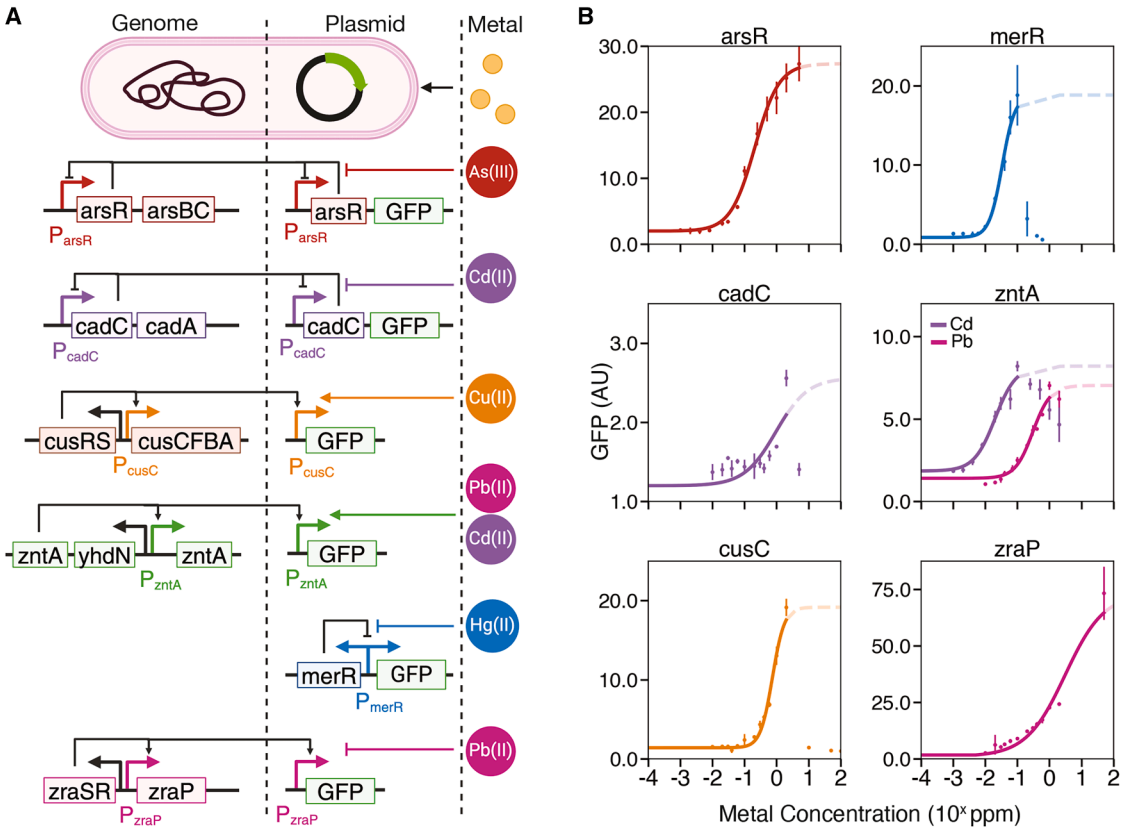



**Figure 1. Development and characterization of real-time transcription factor-based biosensors for heavy metal quantification**

(A) Six plasmids were engineered to respond to As(III), Cd(II), Cu(II), Pb(II), and Hg(II) (rightmost column) in the *E. coli* K-12 MG1655 strain. The center column indicates the engineered circuit elements encoded on the plasmid. In some cases, these engineered elements interact with native transcription factors encoded by the host MG1655 genome, as indicated by the leftmost column. Created with BioRender.com.

(B) Metal-fluorescence dose-response was measured for each of the synthetic fluorescent biosensor strains. Concentrations indicated by the dashed line were excluded from the dose-response curve fit, as they affected cell growth or the GFP response had saturated. Data are mean ± SD, with $n = 3$ independent replicates.

spatially isolated location from the original cell deposit spot, the trapping regions feature back channels that allow for even flow of nutrients to cells through the trap (Figure 2A). Upon device setup, growth media flows through and perpendicular to the spotting region features, carrying spotted cells to the downstream cell traps. Downstream cell traps are loaded as cells flowing through are caught in a 0.5 μm layer. Within roughly 8 h, cell traps reach confluence (Figure S2A). The separation of cell spotting and imaging regions generates distinctive dynamics, enabling integration of multiple reporting mechanisms. Testing GFP as a reporter, we saw a fast, strong readout when induced (Figure 2B). Swapping the fluorescent protein for expression of a cell lysis gene, we demonstrated changes in transmitted light (TL) as another unique mechanism of dynamic response that is compatible with the HD trap geometry and can be used in other biosensing strategies, such as microelectrode-integrated platforms[73] (Figure S3).

To evaluate responses of multiple strains to a single condition in parallel, we utilized a scalable manifold array design capable of housing 48 distinct strains simultaneously, based on our previous multiplexed designs,[7,74–76] enabling assessment of large numbers of strains simultaneously. The 48-strain array maintains a single layer and single inlet-outlet. Past iterations of this multiplexed platform forewent the HD traps, instead opting for a single trap directly connected to the cell-spotting region.[7] While such an implementation generated useful data, HD traps are spatially isolated from the larger cell mass, smaller, and flanked by media channels on both sides, generating clearer, more pronounced dynamics (Figure S2B). Furthermore, direct deposition of cells onto our platform circumvented the need for multiple fluidic ports and cell-loading lines typically required to inoculate multi-strain devices without cross-contamination, greatly simplifying the experimental setup of these devices with manual loading capability.[61,77] This manifold design is easily scalable, allowing for modifications that suit different cell deposition methods (Figures S2C and S2D).

To evaluate a single strain's response to multiple conditions simultaneously, we transposed our modular HD traps into a single-strain gradient generator device for efficient screening across multiple conditions in parallel, based on a previously designed branched mixing array upstream of the cell culture region.[78] A discrete gradient of conditions is generated with distinct mixing ratios of two inlet media (Figure S4A). This design forewent the cell-spotting regions altogether, instead opting for

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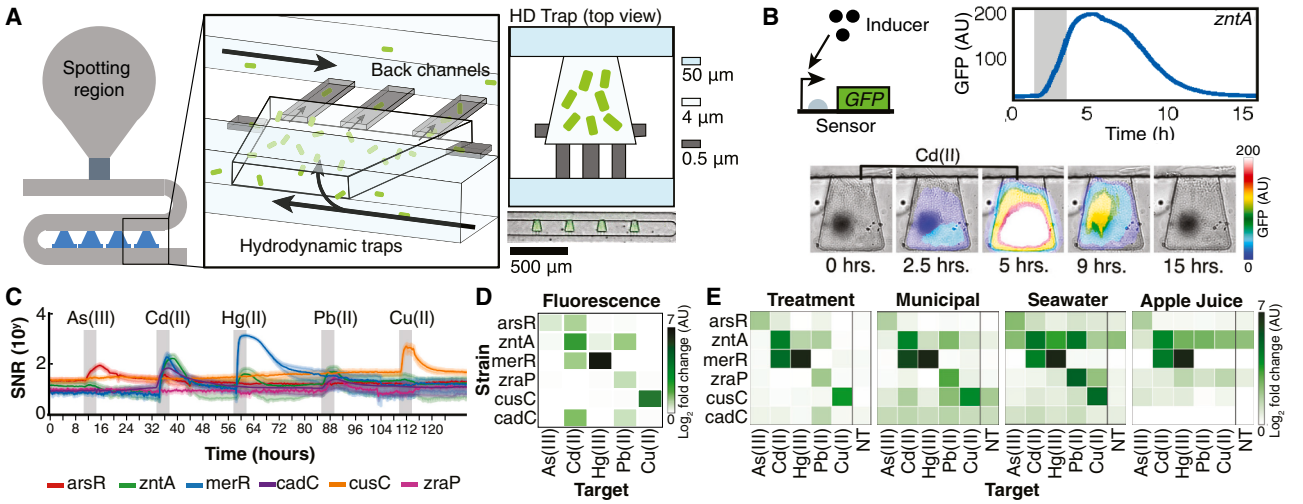



**Figure 2. A versatile, multiplexed microchemostat array**

(A) A drawing of an individual strain's spotting and trapping region. Each growth region is comprised of a spotting region where cells are deposited, after which HD loading enables media flow to carry cells into shallower traps for imaging. Upon filling the traps, media proceed through the main channel, perpendicular to the traps, as indicated by the arrows. The colors in the top view indicate the component heights. Representative trap images are shown.

(B) Inducible fluorescence reporting system (GFP expression) tested with our microfluidic layout. The zntA biosensor is exposed to a 2-h exposure of 1 μM Cd (gray bar). Representative frames are shown; frames in which Cd is present are indicated.

(C) A plot of signal-to-noise ratio of the biosensors subjected to temporal exposures of the five metals, as represented by the gray shaded regions. Fluorescence traces were extracted over time, and signal-to-noise ratio was calculated. Solid lines represent the mean of $n = 12$ HD traps in the microfluidic experiment, and the shaded region represents the standard deviation.

(D) A heatmap of fold-change for each fluorescence biosensor in response to each heavy metal in the experiment described in (C).

(E) The biosensors were tested on four environmental water sources: water from a local water treatment plant, municipal potable water, seawater, and commercial apple juice.

Metal concentrations for (C)–(E) were 1 μM As(III), 0.5 μM Cd(II), 0.5 μM Hg(II), 5 μM Pb(II), and 2 μM Cu(II). No toxin (NT) represents an exposure to the water sample with no additional dissolved metals.

vacuum-loading of cells through the outlet during the final step of the experimental setup. Sulforhodamine dye, an inert dye that does not affect cellular response,[79] was used to visually indicate the gradient across the device as the channels mixed (Figure S4B).

The heavy metal biosensors characterized in batch culture were screened on our multiplexed 48-strain screening platform and subjected to temporal exposures of each of the five metals to characterize their dynamic behaviors (Figures 2C and S5). When exposed to 4-h pulses of each metal, sensors exhibited a typical first-order-like response, increasing fluorescence upon metal introduction, with exponential relaxation upon metal removal. Fluorescence returned to baseline within 24 h of metal removal. The parallel culture of multiple strains on the same device enabled the specific distinction of each metal and the determination of cross-reactivity during each metal exposure (Figures 2C and 2D). Many of the sensors exhibited off-target responses to other non-cognate heavy metals (Figure 2D).

Biosensor response was evaluated on four different water supplies to further evaluate their deployability. The strains were subjected to five consecutive exposures of metals on four different water supplies, spiked with the same heavy metal concentrations as in Figure 2C: water from the Alvarado Water Treatment Plant in San Diego, California, which provides drinking water locally, municipal potable tap water, seawater, and commercial apple juice (Figure 2E). On water from the treatment facility, the activation profiles for all fluorescent sensors were very similar to the results from lab water. However, on municipal potable water, sensors experienced more cross-reactivity, especially the cusC and zntA sensors. Apple juice proved to be a difficult sensing background, especially with its high autofluorescence masking the sensor response (Figure S6). The zntA sensor was continuously activated by apple juice, likely due to the promiscuity of the zntR regulator.[67,80] Furthermore, copper and lead were not detected on this background media by zraP, cusC, and cadC. This may have been attributed to the potential presence of chelating agents or the apple juice's high background fluorescence obscuring the signal. Seawater induced an unconventional response in the sensors, likely due to osmolar stress in the high-salinity environment. Most laboratory strains of *E. coli* used for biosensor development struggle to survive and function under high-salinity environments, suffering hyperosmotic shock and subsequent limitations in population growth.[81] Similarly to its behavior in apple juice, zntA was activated in the absence of any added metal in seawater, yielding a noisy response in the presence of any metal (Figure 2E). In seawater, zntA was better able to distinguish cadmium and lead inducers than in apple juice, but the limit of detection was likely adversely affected due to the noisy response of zntA. Taken together, the observed inconsistency in sensor responses across various non-laboratory background water supplies revealed that the conventional lock-and-key methods of engineering and optimizing biosensors for single-target analytes under laboratory conditions falter when faced with different and often stressful

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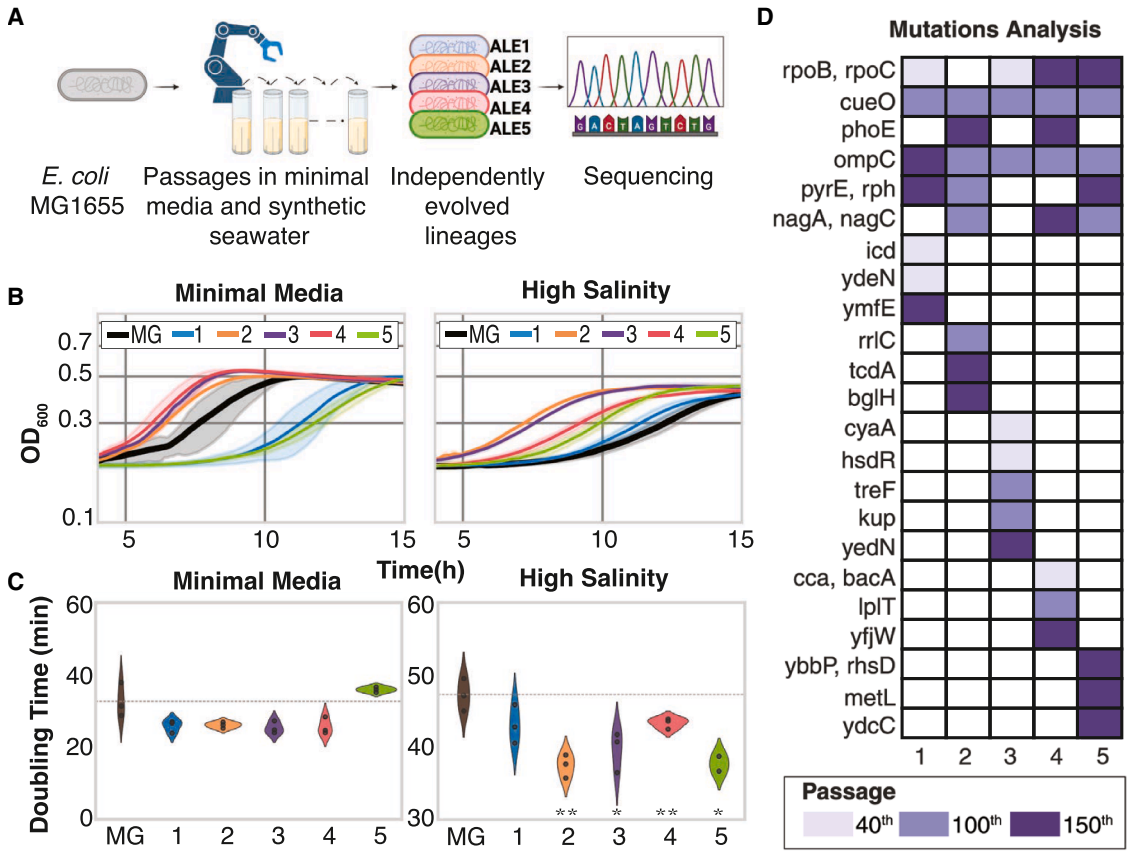



**Figure 3. ALE of *E. coli* K-12 MG1655 to tolerate seawater conditions**

(A) The ALE process. Five independent replicates of *E. coli* K-12 MG1655 were serially propagated in flasks in equal parts minimal media and synthetic seawater (HM9/seawater) using an automated platform. Evolved populations were cryopreserved with every marked increase in growth rate. Final evolved strains, ALE1-ALE5, were isolated at approximately 150 passages and plated on agar. Created with BioRender.com.

(B) Plots of optical density as a proxy for growth for each of the evolved strains in comparison to the starting strain. Solid lines represent the mean of $n = 3$ independent wells, and the shaded region represents the standard deviation. The black line represents the wild-type *E. coli* MG1655 strain.

(C) Doubling times in synthetic seawater are reduced for evolved strains, in comparison to the wild-type strain. $^{*}p = 0.05$ and $^{**}p = 0.01$.

(D) Analysis of genomic changes across different time points of the ALE process for each strain. Shaded boxes represent the presence of a mutation in the identified gene, and the color gradient represents flask number according to the color key.

deployment conditions. Furthermore, disentangling individual signals among combinations of analytes becomes particularly challenging in the presence of cross-reactivity.

## ALE produces diverse, tractable chassis robust in high-salinity environments

We hypothesized that integrating multiple biosensor signals would enhance robustness under deployment conditions where combinations of analytes are present. Although novel chassis using native bacteria are an active area of research for withstanding environmental stressors, genetic tractability and circuit behavior portability remain challenging, hampering efficient engineering for biological applications.[82–84] On the other hand, ALE enables a well-characterized and engineerable strain to efficiently acquire mutations that increase its fitness through prolonged propagation in desirable conditions, such as extreme pH or temperatures.[85–88] We aimed to harness the high-throughput, automated capability of ALE to not only enhance the fitness of *E. coli* K-12 MG1655 in seawater but also to generate genotypically distinct variants with unique phenotypic behaviors without compromising strain tractability or requiring extensive characterization and tuning.

*E. coli* MG1655 has been well-characterized and engineered for many purposes, including increased tolerance against toxins,[89] improved metabolic function,[90] and enhanced detection of pathogens of interest.[91] Previously, MG1655 strains have been evolved to grow rapidly on the non-native substrate sucrose[88] and in minimal media.[87] As such, we evolved the wild-type MG1655 laboratory strain in synthetic seawater to improve its fitness under high-salinity conditions. Using a previously developed automated ALE platform,[87,88] five independent replicates of the MG1655 strain were successively propagated in batch in equal parts minimal media and synthetic seawater (HM9/seawater) to accumulate and select for mutations corresponding to increased fitness in high-salinity conditions. The resulting evolved strains, labeled ALE1-ALE5, were isolated from the final populations and whole-genome sequenced (Figure 3A).

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Evolved strains' growth rates and doubling times in HM9/seawater were estimated using Curveball, an open-source software used for growth curve analysis from Tecan plate reader measurements[92] (Figures 3B, 3C, and S7). Evolved strains exhibited a statistically significant improvement in doubling time in seawater, with at least a 10% reduction below the starting wild-type strain (Figure 3C), suggesting improved fitness to the high-salinity environment presented by seawater. This improvement in growth of the evolved strains over the wild-type strain in high salinity was further validated with NaCl supplementation to minimal media (Figure S8). Furthermore, recent studies have indicated lag phase duration as an organized and adaptive process for bacteria, which can vary upon the presence of stressors.[93] Under high-salinity conditions, the wild-type strain presented a lag phase significantly longer than that of most of its evolved counterparts, suggesting its sensitivity to higher osmolarity[94] (Figures 3B and S9).

Sequence analysis of the evolved strains pointed to various independent mutations that distinguish evolved strains from the wild type (Figures 3D and S10). The *pyrE* and *rph* mutations were commonly associated with ALE growth rate selection in minimal media. *E. coli* K-12 wild types have a frameshift mutation in *rph* causing pyrimidine starvation, which is frequently remedied through the ALE process.[87,95] Apart from these, mutations varied among the independently evolved replicates. Some evolved strains exhibited *phoE* and *ompC* mutations. These are outer membrane porins, which regulate uptake of small molecules.[96,97] To mitigate high-salinity stresses, *E. coli* employ a strategy of uptake or biosynthesis of biocompatible solutes, such as trehalose or L-glutamate, to counterbalance the flood of potassium ions under high osmolarity conditions.[98] As a result, these mutations could be related to biocompatible solute uptake. Studies in the past decade have strongly suggested that adaptation to stressful environments can occur through loss of function, by rewiring the cell's metabolism without the need to refine enzymatic activities or acquire gain-of-function mutations.[99,100] As such, observed mutations such as those in *nagA* and *nagC* may be knockouts that confer stress resistance, as suggested in another recent ALE study with osmotic stress.[101] A knockout study was done with *cueO*, *ompC*, and *nagA* to probe their influences in saline tolerance, but none of these deletions produced notable improvements, suggesting that evolved phenotypes are unlikely to be attributable to single-gene mutations and may depend on more complex adaptations acquired through ALE (Figure S11; Table S2). However, we postulated that the strains' genotypic diversity gives rise to phenotypic diversity among independently evolved strains.

## Dynamic screening of biosensors enhances characterization of response diversity

The six engineered plasmids were transferred into the five newly evolved bacterial chassis variants, producing a library of 30 biosensors. To evaluate their diversity, the biosensors were characterized in our multiplexed microfluidic screening devices to evaluate temporal performance diversity across multiple chassis in HM9/seawater media. Representative temporal fluorescence data for merR-based biosensors collected using the 48-strain microfluidic array are displayed (Figures 4A and 4B). Similarly to the wild-type biosensors, 4-h pulses of each metal yielded first-order-like fluorescence responses before relaxing upon metal removal. However, the evolved strains exhibited diverse dynamic response phenotypes; they varied in several critical parameters, such as the fluorescence fold-change, maximum response rate, and rate of relaxation (Figure 4C). The maximum response rate was computed as the maximum of the first derivative of the fluorescence time series. The response relaxation was analogous to exponential voltage decay in a resistor-capacitor (RC) circuit charge pulse, given by $V_t = V_c e^{-t/\tau}$, where $V_t$ is the voltage across the capacitor at time $t$, $V_c$ is the fully charged state, and $\tau$ is the time constant.[102] The characteristic time constant, $\tau$, is the time in which the signal has decayed 63.2% from its maximum fluorescence, thus allowing it to serve as a proxy for the rate of exponential decay, or rate of signal relaxation. Screening multiple strains simultaneously enabled rapid evaluation of response diversity between evolved biosensors, making desired variant selection efficient (Figure 4D). For instance, if particular biosensing needs required rapid response rates, ALE2 may be desirable, given the faster response rates of the zntA, arsR, and cadC variants. Individual temporal data for the biosensor variants in Figure 4D are available in Figure S12. The diversity in fluorescence fold-change, response rates, and relaxation constants highlighted the phenotypic diversity achieved through the ALE process across evolved variants.

We further demonstrated our modular microfluidic system by using our single-strain gradient generator device for efficient screening of biosensor responses to multiple conditions in parallel. This device was employed to examine the responses of the arsR-MG1655 and arsR-ALE2 biosensors to multiple metal concentrations simultaneously (Figure 4E; Video S1). The two variants were subjected to a gradient of As(III) concentrations automatically generated in the microfluidic device. Sulforhodamine dye was used to visually indicate the 1 ppm As(III)-containing media, as the channels mixed it with plain HM9/seawater media. The temporal GFP responses of the two strains are displayed for each concentration generated by the device (Figure 4F). Using the fold-change responses from the temporal data, the Hill equation was fitted to the responses of arsR-MG1655 and arsR-ALE2 (Figure 4G). The Hill curves revealed that fluorescence response monotonically increased in correspondence with heavy metal inducer concentrations for arsR-ALE2 for nearly double the concentration range of arsR-MG1655, with arsR-MG1655 reaching its maximum at approximately 0.494 ppm and arsR-ALE2 at 0.967 ppm—an improvement in its detection range. The concentrations where a drop-off in fluorescence occurred were omitted from the Hill fit, as the responses were no longer reflective of increasing metal concentration. By using the single-strain gradient device, this difference in monotonic response range was rapidly illuminated.

Beyond GFP fold-change, other dynamic characteristics, including maximum response rate and rate of relaxation, were observed, similarly to the multi-strain device (Figures 4H and 4I). For rate of relaxation, concentrations yielding no significant response (defined as below 1 AU) were excluded. The maximum response rates and rates of relaxation, which varied with induction concentrations, provided additional information beyond fluorescence fold-change that could be useful in biosensor applications. For example, the maximum response rate increased monotonically for the wild-type strain up to approximately 0.6 ppm, despite its fold-change capping at approximately 0.5

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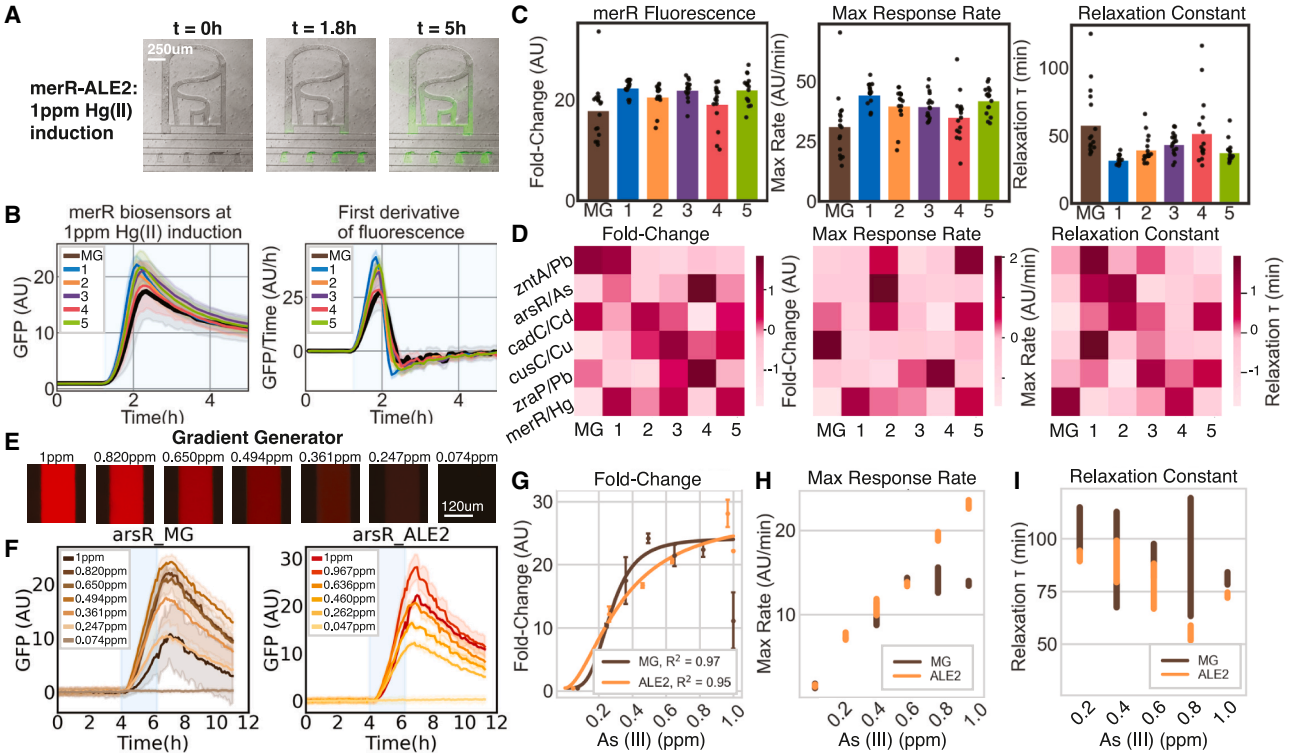



**Figure 4. Evolved strains exhibit diverse dynamic behaviors, including fluorescence fold-change, maximum response rate, and rate of response relaxation in the presence of seawater**

(A) A 48-strain multiplexed microchemostat array was employed to simultaneously induce each evolved strain carrying the same heavy metal-sensing plasmid in HM9/seawater. Representative images of merR-ALE2 induced with 1 ppm Hg(II) in a multi-strain device are displayed.

(B) Fluorescence time traces of each merR biosensor variant, in comparison to merR-MG1655. Solid lines represent mean fluorescence ($n$ = 8 HD traps) measured from HD trap regions for each strain, while shading around lines represents standard deviation. Black line represents the MG1655 biosensor, while all other lines represent evolved biosensor variants. Shading indicates a 4-h induction period.

(C) Maximum fluorescence fold-change of each merR biosensor in comparison to merR-MG1655 (left), maximum response rate of each merR biosensor (center), and relaxation time constant for each merR biosensor (right).

(D) Fluorescence fold-change, maximum response rate, and relaxation time constant for each biosensor in the presence of the indicated heavy metal. Respective units of measure are the same as in (C).

(E) Representative images from microfluidic experiments displaying a gradient generated by sulforhodamine dye (red) as a visual indicator for the presence of heavy metal upon induction.

(F) arsR-MG1655 and arsR-ALE2, exposed to a simultaneous gradient of As(III) concentrations. The blue-shaded box indicates when heavy metal is present. Solid lines represent mean fluorescence ($n$ = 4 HD traps) measured from HD trap regions for each strain, while shading around lines represents standard deviation.

(G) Hill equation fit of arsR-MG1655 and arsR-ALE2 fluorescence fold-change. Data are mean ± SD, with $n$ = 4 independent replicates.

(H) Maximum response rate of arsR-MG1655 and arsR-ALE2 at each As(III) concentration. Data are mean ± SD, with $n$ = 4 independent replicates.

(I) Relaxation time constant of arsR-MG1655 and arsR-ALE2 at each As(III) concentration. Data are mean ± SD, with $n$ = 4 independent replicates.

ppm; therefore, the maximum response rate provided an additional metric that could improve the predictive accuracy of a biosensor beyond single-time-point measurements. Taken together, the multiplexed dynamic characterizations of the WCBs with high temporal resolution demonstrated the response diversity across the evolved strains, achieved with no additional rational engineering. This response diversity could provide more information in the presence of a combination of analytes under deployment conditions than conventional lock-and-key WCBs.

## Rich response diversity across evolved strains facilitates combinatorial prediction

A longstanding limitation of WCBs lies in their difficulty with combinatorial detection, as most samples contain multiple targets, leading to biosensor cross-reactivity and complicating accurate signal interpretation.[50] However, the rich diversity of response phenotypes observed in microfluidics across ALE-generated strains creates the possibility for integrating signals into a fingerprint readout. Instead of relying on lock-and-key specificity, leveraging cross-reactive sensors could enable more reliable and efficient detection of complex analyte mixtures. With the rise of supervised machine learning approaches capable of disentangling multidimensional datasets, this olfactory-inspired, differential sensing approach has become increasingly feasible, expanding the functional potential of bacterial biosensors.[47,55–57] As such, we combined the diverse response phenotypes to train a supervised machine learning regression model to identify combinations of five heavy metals

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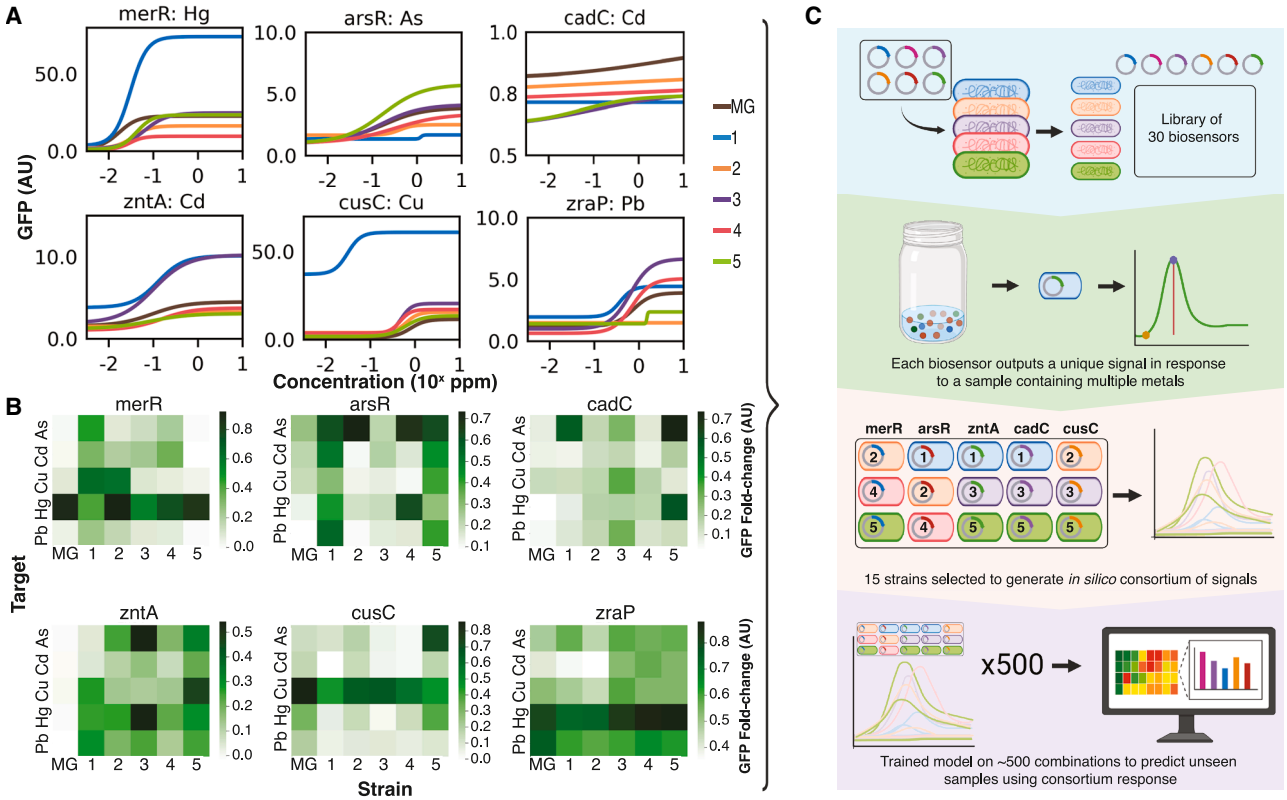



**Figure 5. Response diversity across evolved strains**

(A) Dose-response curves of each evolved biosensor in HM9/seawater when exposed to the heavy metal it was engineered to detect, showcasing the biosensors' diversity. Data shown are the mean, with $n = 3$ independent replicates.

(B) Characterization of specificity across evolved biosensors indicated potential for machine learning classification due to their diversity. Each biosensor, MG1655 and evolved versions, was exposed to each heavy metal. The color gradient corresponds to each biosensor variant's fold-change response to the corresponding metal. Fold changes are min-max normalized by strain to prevent strains with greater fold changes from artificially masking the specificities of strains with smaller fold changes. Heavy metal induction concentrations are the same as in Figures 2C–2E.

(C) Schematic illustrating development of an *in silico* consortium to generate a training set for supervised learning. The library of biosensors was pared down to a set of 15 strains, each of which outputs a signal in response to a sample of mixed heavy metals. Taken together, the consortium of signals constitutes a unique fingerprint readout. These 15 biosensors were exposed to approximately 500 heavy metal combinations, resulting in a rich training set for developing a machine learning model for prediction of unseen samples. Created with BioRender.com.

in seawater samples. By harvesting only the signals, our *in silico* consortium bypasses the need to engineer the strains' coexistence while taking advantage of their diverse phenotypic behaviors. To evaluate its predictive accuracy, we benchmarked the *in silico* consortium's performance against that of conventional single-target WCBs.

As a proof of concept, we used batch culture assays in a Tecan plate reader to characterize features of the evolved biosensors in HM9/seawater. Though these did not capture dynamic features, they were useful in assessing combinatorial capabilities in high-throughput. Characterization of the evolved biosensors' dose-dependent responses revealed distinct response profiles, offering a diverse library of biosensors that could be selected based on desired detection ranges or sensitivities to meet specific biosensing needs (Figures 5A and S13).

Furthermore, we examined each biosensor's fold-change responses to the panel of heavy metals in HM9/seawater to evaluate cross-reactivity (Figure 5B). Evolved strains harboring identical plasmids displayed distinct specificity and cross-reactivity patterns. In fact, some biosensors exhibited greater responses to non-cognate heavy metals than to their intended targets. For example, merR and arsR in ALE1 exhibited broadened responsiveness across multiple metals. Most zntA-based biosensors remained promiscuous, similar to the wild-type zntA biosensor, likely because it is a P-type ATPase that pumps multiple divalent heavy metals, including lead, cadmium, and zinc.[103,104] Some evolved strains carrying cadC, which exhibited weak specificity and sensitivity to cadmium despite reports in literature,[63] showed enhanced responses to non-cognate metals, such as mercury and arsenic, as in ALE5. Meanwhile, zraP-based biosensors displayed strong specificity to mercury and high cross-reactivity to other heavy metal inducers.

Instead of treating cross-reactivity as a hindrance as in conventional single-target WCBs,[49,50] we leveraged it by integrating the phenotypic responses of multiple evolved biosensors to build our *in silico* consortium. This approach generated distinctive response signatures—analogous to barcodes—that circumvented cross-reactivity and enabled more robust quantifications

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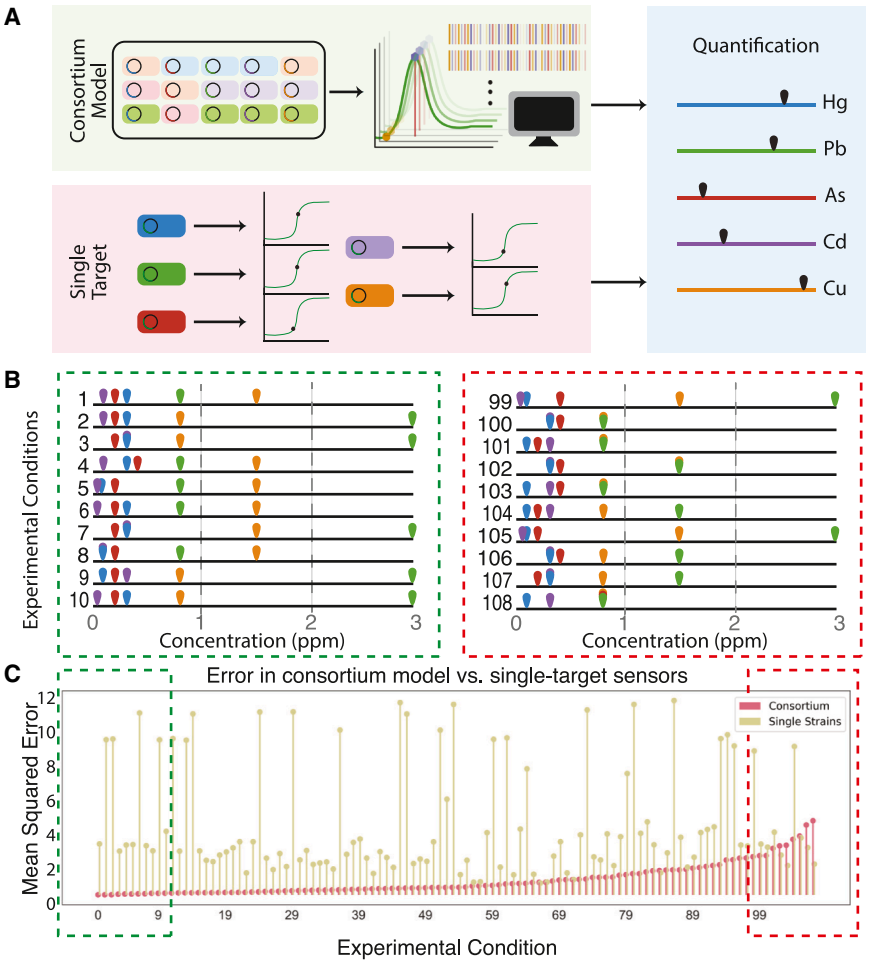



**Figure 6. Consortium model outperforms single-target WCBs in predictions of heavy metal combinations**

(A) Schematic illustrating consortium biosensor model prediction benchmarking against conventional single-target WCB dose-response quantification.

(B) Sample experimental conditions, previously unseen by the model, each of which consists of a combination of all 5 heavy metals. Conditions in which the model performed the best and the worst are shown, and concentrations of each heavy metal in the test conditions shown are indicated.

(C) Total MSE between predicted and true values across heavy metals in each experimental condition, for both model and single-target WCB predictions. Organized in order of increasing error in consortium model predictions, MSE values closer to zero indicate reduced error between predicted and true values. Conditions in which the model performed the best and worst are highlighted in (B).

of individual metals in mixed samples using machine learning (Figure 5C). For each sensor construct, all evolved strain specificities were ranked from highest to lowest, irrespective of the original target metal. The top three evolved strains were selected for each sensor construct, under the assumption that strong specificity, even if redirected, could enhance discriminatory capability. Several biosensors exhibited secondary specificity, such as arsR-ALE4, or even tertiary specificity, such as cadC-ALE5. zraP-based biosensors were excluded due to their pronounced responses to mercury and broad promiscuity, given that most merR-based biosensors already demonstrated strong mercury specificity. The final selection of 15 strains used for consortium model development is indicated in Figure 5C.

Using an acoustic droplet handler, approximately 500 unique heavy metal combinations were introduced to each of the 15 selected biosensors to generate high-throughput training data. The combinations spanned varying concentrations of the five heavy metals, each within the biosensors' operational ranges. Features were extracted from each biosensor's profile for proof of concept, including fluorescence fold-change maxima, minima, and curve skewness, to generate a distinct response signature for each combination (Figure S14A). Supervised machine learning regression models were trained on these response signatures to predict the concentrations of individual heavy metals present in each combination. Multiple regression approaches were implemented and compared, including polynomial regressions, decision trees, and ordinary least squares (OLSs), using the scikit-learn Python package.[105] Model performance was benchmarked using the sum of mean squared errors (MSEs) between predicted and true metal concentrations in each combination (Figure S14B). To prevent bias and overfitting, we used $k$-fold cross-validation with $k = 6$ randomly grouped segments, where each segment, k, served as a validation set while the model was trained on the remaining segments. The MSE was averaged across the combinations and $k$-folds to evaluate predictive accuracy. The OLS model proved sufficient. A common criticism of complex machine learning models is their "black box" nature, obscuring how the predictions are derived.[106] However, the OLS model provides interpretability by revealing the relative magnitudes of its coefficients, providing insight into biosensor signal contributions to metal concentration predictions (Figure S14C).

To evaluate the predictive performance of the model, an experimental test dataset comprising over 100 previously unseen heavy metal combinations was generated, and the trained model was applied to this test set. The consortium model's performance was benchmarked against the conventional single-target WCB paradigm (Figure 6A). For comparison, an optimal WCB for each of the five heavy metals was identified as the biosensor yielding the lowest MSE in predicting its intended target across all training combinations using its dose-response data (Figure S15). Using the same MSE metric, the consortium model predictions were benchmarked against the single-target WCB predictions for each heavy metal in every experimental condition. The consortium-based predictions outperformed the single-target WCB predictions in 93% of the test combinations,

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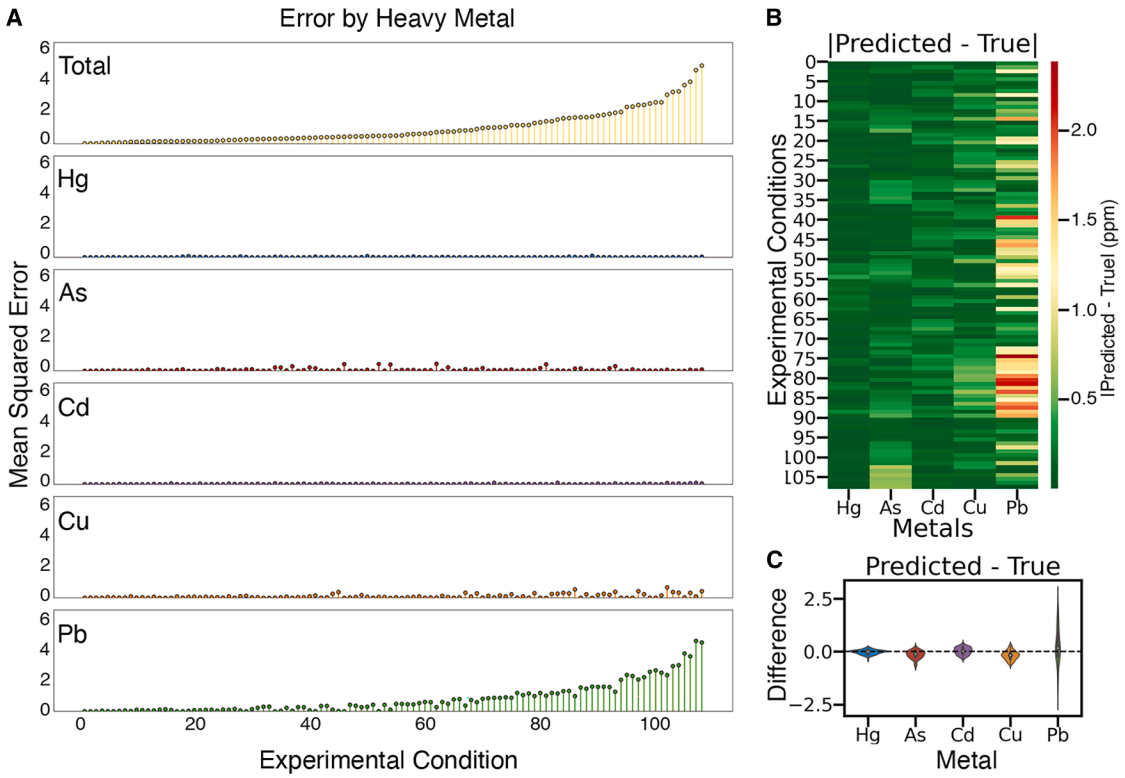



**Figure 7. Analysis of error identifies where the consortium model needs improvement**

(A) Total MSE of consortium model predictions, disaggregated by each individual heavy metal. Experimental conditions are the same as in Figures 6B and 6C.

(B) Heatmap of absolute difference between predicted and true heavy metal concentrations across experimental conditions for each heavy metal.

(C) Violin plots, by heavy metal, of the difference between predicted and true values.

exhibiting reduced error when leveraging the collective responses (Figures 6B, 6C, and S16). We further analyzed the consortium model-based predictions with those of the single-target WCBs for each heavy metal constituent (Figure S17). Across most metals, with the exception of cadmium, the consortium model demonstrated superior performance in a majority of the test combinations: 65% for mercury, 76% for lead, 92% for arsenic, 49% for cadmium, and 82% for copper. Finally, we assessed false negative rates, as failing to detect the presence of a metal can have critical consequences. Besides mercury, the consortium model consistently reduced false negatives relative to single-target WCBs (Figure S18).

To further evaluate the consortium predictions, we separated the total error into its individual heavy metal constituents to analyze each metal's error contributions (Figure 7A). This deconvolution revealed that imprecision in lead concentration predictions consistently contributed a majority of the error in the conditions where the model produced greater error. This is further exemplified in the absolute differences between the predicted and true concentrations across the experimental conditions (Figures 7B and 7C). With the exception of lead, the consortium model predictions were mostly within 0.5 ppm of the true concentrations. Mercury and cadmium were the most closely predicted, with arsenic and copper consistently slightly underestimated. Lead predictions were vastly varied, which is likely due to the promiscuity of zntA, the primary lead sensor in the consortium. Further improvement of the consortium model could potentially be achieved by the inclusion of more WCBs with diverse responses to lead, such as the excluded zraP biosensors, or tuning the lead sensor plasmid construct itself to produce more specific responses.

## DISCUSSION

As whole-cell bacterial biosensor development has focused on detecting targets in isolation under controlled laboratory conditions, we developed an *in silico* consortium model paradigm to combinatorially quantify multiple heavy metals in seawater. We employed ALE to obtain a diverse range of *E. coli* strains equivalently optimized for high-salinity seawater environments without additional rational engineering. Housing transcription factor-based biosensing constructs for the detection of five heavy metals relevant to seawater in these new variants, the final WCBs were screened using a high-throughput, multiplexed, modular microfluidic platform for continuous culture, highlighting the diversity achieved through the ALE process. Leveraging this diversity as a "fingerprint" readout, we used supervised machine learning regression to quantify combinations of heavy metals in seawater conditions. Toward environmental deployment and monitoring, the tools developed in this work represent a stepping stone in approaching biosensor engineering that broadens the deployable potential of previously developed biosensors while

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bypassing the need to improve their specificity or sensitivity to overcome off-target effects. Furthermore, this paradigm shift of using a collective response rather than specific lock-and-key responses could pave the way for detecting targets that currently have no developed single-target biosensors. While our approach presents a unique solution toward the biosensing space, limitations and challenges still exist to solidify its utility.

We demonstrate the use of ALE for efficiently achieving resilient variants for conditions of interest.[85–88] Although our study focused on adapting strains for high-salinity conditions of seawater, a multitude of stressors exist in natural environments, including ranges of temperature, pH, oxygen, or even the sensing target itself, which we observed for some heavy metals at higher concentrations, such as mercury. These stressors present operational limitations, such as limiting the range of concentrations that can be reliably quantified. Future work in ALE for biosensors should aim to develop strains robust to these conditions and to broaden the range of detectable concentrations beyond current sensor limitations due to target toxicity. With bioinformatic capabilities, analyzing resulting strains' genomic mutations and probing their effects on fitness in specific environments can inform future rational engineering approaches.[85–87]

Multiplexed microfluidic platforms fully utilize the diverse sensing target capabilities of WCBs, enabling the screening of multiple engineered strains under multiple conditions while collecting their dynamic data in real time.[77,107] We believe the accessibility, throughput, dynamics, and week-long time scales achievable with our microfluidics screening tool can push the frontier of the biosensing landscape. More broadly, the microfluidics platform allows for parallel screening for desired phenotypes, a persistent challenge in synthetic biology design, as our generation of genotypic diversity exceeds our ability to screen complex phenotypes.[108] While not a perfect recreation of complex real-world environments, multiplexed microfluidic capabilities for tuning environmental and time-dependent parameters can serve an important role in the prototyping and scale-up of all classes of gene circuits, including oscillators, logic gates, and feedback controllers, which have been increasingly deployed in time-dependent applications spanning metabolic engineering to therapeutic delivery.[18,109–112]

Currently limited to the collection of training data, the model could be rapidly refined with improved high-throughput data collection approaches or systematic methods for collecting and sharing training data among the biosensing community. Further precision can also be achieved with tuning individual biosensors for specificity through engineering the sensor or transducer components or through evolutionary or computational approaches.[40–45] The biosensor collection itself may also be further refined. Although a subset of 15 biosensors was selected for this study as a proof of concept, smaller subsets could be subjected to the regression analysis, scaling up until performance achieves diminishing returns, potentially narrowing down a set of biosensors most useful in decision-making or perhaps even identifying those doing more harm than good to quantification. If such a set of essential biosensors exists, analyzing their features and their contributions to the model could be informative of the diversity of behaviors most useful in biosensor design. Furthermore, as scalable training and validation pipelines are developed, extending analysis of consortium model performance to a broader range of environmental conditions would be valuable in refining the approach.

Lastly, future approaches may consider other machine learning models. Other pattern recognition machine learning models that have been employed in similar array-based or combinatorial approaches to detect a unique fingerprint output include *k*-nearest neighbors, random forests, *k*-means clustering, and gradient-boosted trees.[47,55,56] These have been used in artificial nose and tongue applications, which, inspired by natural olfactory senses, similarly aim to disentangle multiple inputs into a single output. Linear discriminant analysis (LDA), principal-component analysis (PCA), support vector machine (SVM) regression, and hierarchical agglomerative clustering (HAC) are great tools for pattern recognition that can be incorporated into the framework of biosensor engineering and design.[47] Considering the output of diverse signals with the intention to generate distinguishable patterns that work well with these machine learning algorithms may create more degrees of freedom for future biosensor design and engineering than limiting it to increasing sensitivity and specificity of single-target lock-and-key biosensors.

## RESOURCE AVAILABILITY

### Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Alyssa Jasmine Chiang (a3chiang@ucsd.edu).

### Materials availability

This study did not generate any new materials.

### Data and code availability

All data have been deposited at Zenodo and are publicly available as of the date of publication. DOIs are listed in the key resources table. All original code has been deposited at Zenodo and is publicly available as of the date of publication. DOIs are listed in the key resources table. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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## AUTHOR CONTRIBUTIONS

A.J.C. and J.H. designed the project. A.J.C., M.F., N.H., K.Z., and J.H. wrote the paper. A.M.F. designed the ALE experiments and interpreted the results. E.C.O. and R.M. executed the ALE experiments. N.C., R.O., and E.S. designed and tested the multi-strain microfluidic screening devices. A.J.C. and A.D. designed the single-strain gradient microfluidic device. L.B. designed the biosensor plasmids. P.N. provided guidance for implementing the supervised

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machine learning regression. A.J.C. and M.A. performed and analyzed the evolved biosensor and high-throughput experiments.

## DECLARATION OF INTERESTS

J.H. declares that he is a co-founder of GenCirq, which focuses on cancer therapeutics. He is on the board of directors and has equity in GenCirq. His spouse is employed part time by GenCirq for bookkeeping and employee support with human resources.

## STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS
- METHOD DETAILS
  - Biosensor plasmid construction
  - Plate reader experiments
  - Microfluidic device development and fabrication
  - Microfluidic device loading and bonding
  - Microfluidic experimental protocol
  - Source water testing
  - Adaptive laboratory evolution
  - Knockouts
  - Batch growth assays
  - Supervised learning regression
- QUANTIFICATION AND STATISTICAL ANALYSIS

## SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.cels.2026.101702.

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## STAR★METHODS

### KEY RESOURCES TABLE

| REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|
| Bacterial and virus strains | | |
| *E. coli* strain MG1655 | ATCC | 47076 |
| Chemicals, peptides, and recombinant proteins | | |
| Gibson Assembly Master Mix | New England Biolabs | E2611S |
| Sylgard 184 silicone elastomer | Dow | 1317318 |
| Mercury | Waters ERA | ID 027 |
| Arsenic | Waters ERA | ID 013 |
| Lead | Waters ERA | ID 024 |
| Cadmium | Waters ERA | N/A |
| Copper (II) Sulfate Pentahydrate | Sigma-Aldrich | CAS 203165 |
| Artificial Seawater | Cole-Palmer | CAS 8363-1 |
| LB Broth | BD | DF0446-07-5 |
| Kanamycin | Sigma-Aldrich | 25389-94-0 |
| Tween-20 | bioWORLD | CAS 9005-64-5 |
| Critical commercial assays | | |
| Monarch Plasmid Miniprep Kit | New England Biolabs | N/A |
| Oligonucleotides | | |
| Primers | Integrated DNA Technologies, San Diego, CA | N/A |
| Software and algorithms | | |
| Sci-kit learn | https://scikit-learn.org/stable/ | N/A |
| Curveball | https://curveball.yoavram.com/ | N/A |
| Original code | This paper | Zenodo: https://doi.org/10.5281/zenodo.19361653 |
| Other | | |
| Sanger sequencing | Integrated DNA Technologies, San Diego, CA | N/A |
| Spectrophotometer | N/A | N/A |
| Plates | N/A | N/A |
| Echo Acoustic Liquid Handler | Beckman Coulter, Inc., Indianapolis, IN | N/A |
| Infinite® M200 PRO Multimode, Microplate Reader | Tecan, Switzerland | N/A |
| Nikon TE2000-U Epifluorescent Inverted Microscope | Nikon Instruments Inc., Tokyo, Japan | N/A |

### EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

*E. coli* MG1655 was used as the primary strain in all experiments. Adaptive laboratory evolution was conducted on five biological replicates of *E. coli* strain MG1655 for the experimental evolution protocol.

### METHOD DETAILS

#### Biosensor plasmid construction

Biosensing plasmids were constructed using PCR amplification of host strain components listed in Table S1, followed by Gibson Assembly. The backbones containing green fluorescent protein were obtained from existing plasmids in our lab. Plasmid sequences were confirmed with Sanger sequencing (Integrated DNA Technologies, San Diego, CA) and transformed into chemically competent *E. coli* K-12 MG1655 cells and evolved variants.

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### Plate reader experiments

Overnight cultures grown in LB media were reseeded into HM9 minimal media or 1:1 HM9 to synthetic seawater (HM9/seawater) at 1:1000 dilution and grown until they reached an $OD_{600}$ of about 0.3-0.4. The HM9 minimal media composition, containing minimal traces of metal, was based on prior studies[7,113] and optimized for microfluidic *E. coli* growth. For fluorescence dose-response curves, 96-well plates were prepared with media and varying heavy metal concentrations. Cultures were diluted to $OD_{600}$ of about 0.1 on the plate. Fluorescence and $OD_{600}$ values were measured using an Infinite® M200 PRO Multimode Microplate Reader. For all plate reader experiments, fluorescence measurements were taken at maximum induction, which occurred around 2 hours into the experiment. This was normalized by fluorescence pre-induction to correct for cell autofluorescence and by $OD_{600}$. For characterizing response specificity across evolved strains, responses were min-max normalized, to prevent any strain's responses from obscuring the others, allowing visualization of specificity.

### Microfluidic device development and fabrication

Microfabrication techniques in patterning SU-8 photoresist onto a silicon wafer to create a mold for microfluidic devices have previously been described.[71,72,75] Polydimethylsiloxane (PDMS) microfluidic devices were made with a mixture of 10:1 Sylgard 184 to curing agent, which was poured over the wafer centered on a glass petri dish and degassed to remove bubbles in the PDMS. The wafer with the PDMS was cured for one hour at 95°C on a level surface.

### Microfluidic device loading and bonding

PDMS devices and glass slides were cleaned with MilliQ and 70% Ethanol. Adhesive tape was used to remove any remaining fibers on the device. Overnight cultures were reseeded at 1:100 to an $OD_{600}$ of approximately 0.3-0.4. For multi-strain devices, 1mL of the overnight cultures was centrifuged, the supernatant was discarded, and cell pellets were resuspended in 20 μL fresh LB to make highly concentrated cell cultures. The devices and slides were exposed to oxygen plasma, after which 23 ga syringe tips were used to transfer cells from the concentrated suspension to the devices by lightly contacting the cell traps on the PDMS. The device and glass slide were then bonded and cured at 37°C for two hours. To load the gradient devices, they were instead bonded and cured prior to vacuum loading using a previously described protocol.[78]

### Microfluidic experimental protocol

Microfludic experiments were performed on a Nikon TE2000-U epifluorescent inverted microscope (Nikon Instruments Inc., Tokyo, Japan). The inlet and outlet ports were connected to 50mL syringes and Tygon tubing. The inlet syringes were filled with HM9/seawater with Kanamycin and 0.075% Tween-20. Tween-20 is a surfactant to help prevent biofilm formation that can clog microfluidic channels and reduce the longevity of experiments. This has previously been shown to have no adverse effect on *E. coli* in many microfluidic experiments.[72,114] The waste outlet was filled with approximately 5 mL of MilliQ water, to provide negative pressure. The inlet and outlet syringes were separated by a 10" height difference. Strains were grown to confluence in the cell traps after at least 8 hours, and a predetermined heavy metal concentration was pipetted into the inlet syringe. After a 4-hour induction period, any remaining inducer solution was pipetted out and replaced with fresh HM9/seawater. Fluorescence time series were extracted and normalized to remove device background fluorescence and strain background fluorescence from pre-induction levels. Maximum fluorescence measurements were taken at maximum induction. Maximum response rates were taken at the maximum of the first derivative of the temporal fluorescence response. Lastly, relaxation time constants were measured from the time point of the maximum response to 4 hours after.

### Source water testing

Sensor strains were tested on four different water sources: (1) water from the Alvarado Water Treatment Plant in San Diego, CA, (2) municipal potable tap water from San Diego, CA, (3) Seawater from Blacks Beach in La Jolla, CA (3) and apple juice (Trader Joe's Fresh Pressed Apple Juice SKU#88463). Once a day, each device was exposed to a mixture of a single toxin dissolved in HM9 made from the respective water source for four hours. Signal-to-noise ratio was determined as the fluorescence value at each time point divided by the standard deviation of fluorescence 7.5 hours before induction. Fold change was calculated from the signal-to-noise ratio for four biological replicates of each sensor and normalized by fluorescence of a promotorless strain to correct for autofluorescence.

### Adaptive laboratory evolution

Adaptive laboratory evolution (ALE) was carried out by Adam Feist using a previously described method.[87,88] Briefly, five biological replicates from *E. coli* strain K-12 MG1655 were grown overnight in HM9/seawater. Using an automated liquid-handler, this culture was serially passaged during exponential phase over approximately 140 passages into fresh HM9/seawater. Final strains used in experiments were isolated at the end of the evolution process and whole-genome sequenced.

### Knockouts

Knockouts were carried out using λ Red as described in Datsenko and Wanner,[115] with primers included in Table S2.

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### Batch growth assays

Overnight cultures were grown in LB media at 37°C with antibiotics for 16 hours and reseeded into 1:100 diluted HM9/seawater media. Once an $OD_{600}$ of approximately 0.3-0.4 was reached as measured by the Spectrophotometer, they were diluted to $OD_{600}$ of approximately 0.01. Growth was then monitored by measuring $OD_{600}$ were taken every 5 minutes for 12-16 hours.

### Supervised learning regression

The training and test sets were generated by selecting metal concentrations that lay within sensor operational ranges for each metal, and combinations were computationally produced. The corresponding amounts were transferred to 384-well plates using the Labcyte Echo 550 Acoustic Liquid Handler, which is capable of transferring at the nanoliter scale. Each selected strain was exposed to every condition. The scikit-learn machine learning package in Python was used for training and testing the different regression models, including polynomial regression, ordinary least squares (OLS), and decision trees. Features were engineered by normalizing each strain's GFP fluorescence to fluorescence with no inducer added. Maximum fluorescence, minimum fluorescence, and skewness of curve were extracted for each strain. Once standardized to prevent feature bias, these three features across the 15 strains served as input (total of 45 features) yielding predictions of each heavy metal concentration as output. To train the model to recognize the outputs, k-fold cross validation was used with k=6 segments. Mean squared error was computed between the model's predicted outputs and the true metal concentrations.

## QUANTIFICATION AND STATISTICAL ANALYSIS

To calculate the normalized fluorescence of multi-strain microfluidic experiments, the mean fluorescence pixel value (GFP) for each trap was extracted across the 10X microscope image stack using ImageJ. All GFP values were background-normalized to the mean GFP pixel value in a selection of equal size in a part of the chip with no cells ($GFP_{BG}$). Background-normalized GFP values were normalized to the inverse of similarly background-normalized transmitted light $\left(\frac{TL}{TL_{BG}}\right)$ pixel values, as a proxy for cell density. The following formula was used:

$$\text{Normalized Fluorescence} = \frac{GFP}{GFP_{BG}} \times \frac{TL_{BG}}{TL}$$